A PROJECT TITLE
ON
CARBON EMISSION PREDICTION USING GRADIENT BOOSTING AND LINEAR REGRESSION

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
In
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By

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Abstract

This project presents the design, implementation, and evaluation of a machine learning system for carbon emission prediction. The study covers dataset collection, data preprocessing, exploratory data analysis, feature selection, model development, training, testing, and performance evaluation. Two machine learning algorithms, Linear Regression and Gradient Boosting, are used to predict carbon emission levels based on available historical and environmental data. The models help identify patterns and relationships between different factors influencing carbon emissions. The predicted results can be used to understand emission trends and support environmental planning and sustainable decision-making. The project also highlights the potential of machine learning in addressing environmental challenges and promoting a cleaner and greener future.

Keywords: Carbon Emission Prediction, Machine Learning, Linear Regression, Gradient Boosting, Environmental Data Analysis, Sustainable Development.Keywords:

1. Introduction

1.1 Purpose of the Project

Carbon emissions are one of the major contributors to global warming and climate change. Activities such as electricity generation, transportation, industrial production, and the consumption of fossil fuels release carbon dioxide into the atmosphere. Understanding how these emissions change over time is important for environmental management, climate planning, and sustainable development. Traditional methods of estimating emissions may depend on statistical calculations and historical trends, which can have limitations when relationships between influencing factors are complex. Machine learning provides an opportunity to analyze historical data, identify patterns, and estimate carbon emission levels using data-driven models.

The purpose of this project is to develop a machine learning-based system that predicts carbon emission levels using Linear Regression and Gradient Boosting. The project aims to compare a simple regression approach with an ensemble learning technique that can model more complex relationships in data. The intended stakeholders include environmental researchers, policymakers, industries, and organizations interested in monitoring emissions and planning sustainability initiatives. The resulting predictions can support environmental analysis and decision-making, although their usefulness depends on the quality of the dataset and the accuracy of the trained models.

1.2 Problem Statement

The objective of this project is to develop a regression-based machine learning system that predicts carbon emission values from relevant historical, environmental, or energy-related input features. The dataset is prepared through data cleaning, missing-value treatment, exploratory data analysis, and feature selection, after which Linear Regression and Gradient Boosting Regression models are trained and tested. Their performance is compared using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (\(R^2\)). The expected output is a numerical estimate of carbon emissions for given input conditions and an evidence-based comparison of the two algorithms.

1.3 Existing Methods and Disadvantages

Traditional carbon emission estimation methods include statistical regression, emission-factor calculations, and time-series forecasting. Linear statistical models are useful for identifying relationships between input variables and emission levels. However, a simple linear model may not adequately capture nonlinear relationships, interactions among multiple factors, or sudden changes in emission patterns. Its performance also depends on the quality of the input data, the suitability of its assumptions, and the representativeness of the observations used for training. Research reviews have examined a wide range of carbon-emission prediction models and identified differences in their predictive capabilities and applications.

More advanced approaches include neural networks, support vector regression, random forests, boosting algorithms, and hybrid prediction models. These techniques can capture more complicated relationships but may require greater computational resources, parameter tuning, and careful validation. Complex models may also overfit the training data, while models trained on one location or period may not generalize well to another. Therefore, evaluating models on unseen data and selecting suitable evaluation metrics are essential for determining whether a model is practically useful

1.4 Proposed System

The proposed system uses a machine learning pipeline to predict carbon emissions from a suitable dataset. Initially, the dataset is inspected and preprocessed to handle missing values, duplicate records, inconsistent data, and relevant categorical features. Exploratory data analysis is then performed to understand feature distributions and relationships with the target variable. The prepared dataset is divided into training and testing subsets, and Linear Regression and Gradient Boosting Regression models are developed. The models are evaluated using MAE, MSE, RMSE, and \(R^2\), and their results are compared to identify the more suitable approach for the selected dataset. The proposed system aims to provide a clear, reproducible comparison of a baseline regression model and a more flexible ensemble model for carbon emission prediction.

1.5 Research Objectives(atleast 4)

1. To identify and prepare a suitable carbon emission dataset for regression analysis.

2. To perform data preprocessing and exploratory data analysis to understand the dataset.

3. To identify relevant input features while preventing data leakage.

4. To develop and train Linear Regression and Gradient Boosting Regression models.

5. To evaluate and compare model performance using MAE, MSE, RMSE, and (R^2).

6. To determine the better-performing model based on experimental results.

7. To analyze the limitations and practical applicability of the developed prediction system.

2. Literature Survey (recent studies from 2-3 years)

Review ten recent and directly relevant studies. Verify bibliographic details, DOI, volume, issue, pages/article number, and year before final submission.

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	A Review of Macroscopic Carbon Emission Prediction Model Based on Machine Learning (2023). Sustainability, 15(8), 6876. DOI: 10.3390/su15086876	Zhao, Y. et al. (2023). Sustainability, 15(8), 6876. DOI: 10.3390/su15086876	Traditional prediction methods may struggle with nonlinear emission patterns and changing environmental and economic factors.	To review and compare machine-learning models for predicting large-scale carbon emissions and identify their advantages and limitations.	Literature review of Backpropagation Neural Networks (BPNN), Support Vector Machines (SVM), Long Short-Term Memory (LSTM), Random Forest (RF), and Extreme Learning Machines (ELM).	Machine-learning methods offer useful approaches to carbon-emission forecasting, but their suitability depends on the data and prediction problem.	Compare more algorithms using common datasets, improve model generalization, and explore hybrid models and additional influencing factors.
2	Development and Performance Comparison of Optimized Machine Learning-Based Regression Models for Predicting Energy-Related Carbon Dioxide Emissions (2023)	Akkaya, E. K. and Akkaya, A. V. (2023). Environmental Science and Pollution Research, 30, 122381–122392. DOI: 10.1007/s11356-023-30955-1	Accurate prediction of national CO_2 emissions is difficult because emissions depend on multiple factors and differ between countries.	To develop and compare machine-learning regression models for predicting CO_2 emissions across 68 countries.	Nine regression models using Support Vector Regression, ensemble trees, and Gaussian Process Regression; Bayesian optimization was used to tune hyperparameters.	The optimized Gaussian Process Regression model performed best in the study, with \(R^2=0.9998\), MAE = 4.904, and RMSE = 10.328 on its test data.	Validate models on additional countries and independent datasets, and examine performance when economic and energy conditions change.
3	Industrial Carbon Emission Forecasting Considering External Factors Based on Linear and Machine Learning Models (2024)	P. and Wang, S. (2024). Journal of Cleaner Production, 434, 140010. DOI: 10.1016/j.jclepro.2023.140010	Previous studies often focus on large-scale emissions, while urban-level forecasting and the influence of external factors receive less attention.	To build a forecasting framework that accounts for temporal changes and economic, industrial, demographic, and urban-development factors.	Twelve initial influencing factors; Lasso regression for feature selection; a combined ARIMA–Support Vector Regression model for linear and nonlinear patterns.	The proposed framework reported \(R^2=97.5\%\) and MAPE = 0.096 in the study, and it was used to examine six future development scenarios.	Test the framework in more cities, investigate additional external factors, and evaluate predictions under different environmental and industrial policies.

4	Day Ahead Carbon Emission Forecasting of the Regional National Electricity Market Using Machine Learning Methods (2023)	Engineering Applications of Artificial Intelligence, 123, Part B, 106314 (2023). DOI: 10.1016/j.engappai.2023.106314	Electricity-grid emissions vary with regional electricity generation and the proportion of renewable and non-renewable energy.	To forecast next-day carbon-emission intensity across regions of Australia's National Electricity Market.	Machine-learning forecasting models, including a particle-swarm-optimized Extremely Randomized Trees model, compared with LSTM and Extreme Learning Machines; five regional datasets were released.	The optimized tree model outperformed the compared models in the reported study. Forecasting performance varied with the renewable-to-non-renewable generation ratio.	Incorporate changing weather and electricity demand, test performance across longer periods, and assess transferability to other electricity markets.
5	Application of Neural Networks on Carbon Emission Prediction: A Systematic Review and Comparison (2024)	Energies, 17(7), 1628 (2024). DOI: 10.3390/en17071628	Neural-network approaches differ in architecture and performance, making it difficult to select a suitable method for carbon-emission prediction.	To systematically review neural-network methods used in carbon-emission prediction and compare their applications.	Systematic review and comparison of neural-network approaches for emission prediction.	Neural networks provide flexible methods for learning complex relationships, but performance depends on the model design, data, and evaluation process.	Use standardized evaluation datasets, improve model interpretability, and compare neural networks against conventional regression and ensemble methods.
6	Forecasting CO_2 Emissions of Fuel Vehicles for an Ecological World Using Ensemble Learning, Machine Learning, and Deep Learning Models (2024)	Gurcan, F. (2024). PeerJ Computer Science. DOI: 10.7717/peerj-cs.2234	Estimating vehicle CO_2 emissions accurately is important for assessing environmental impacts and supporting emission-reduction efforts.	To compare a broad range of machine-learning, ensemble-learning, and deep-learning regression algorithms for predicting vehicle emissions.	Comparative regression analysis using 18 algorithms from machine-learning, ensemble-learning, and deep-learning approaches.	The study provides a broad comparison of regression methods for vehicle emissions and demonstrates the value of evaluating multiple algorithms rather than relying on one model.	Validate models on different vehicle types and driving conditions, investigate feature importance, and compare predictive performance with computational cost.
7	The Development of a Machine Learning-Based Carbon Emission Prediction Method for a Multi-Fuel-Propelled Smart Ship by Using Onboard Measurement Data (2024)	Sustainability, 16(6), 2381 (2024). DOI: 10.3390/su16062381	Ships using different fuels and operating under changing conditions require reliable ways to estimate their carbon emissions from actual measurements.	To develop a machine-learning approach for predicting emissions from a multi-fuel-propelled smart ship using onboard data.	Machine-learning prediction based on onboard measurement data from a multi-fuel ship.	The work demonstrates the application of data-driven prediction to ship emissions, where operating measurements can help capture complex relationships.	Test the method across different vessels, fuel combinations, and operating conditions, and evaluate its performance with longer-term real-world measurements.

8	CNN-GRU-Attention Neural Networks for Carbon Emission Prediction of Transportation in Jiangsu Province (2024)	Sustainability, 16(19), 8553 (2024). DOI: 10.3390/su16198553	Transportation emissions change over time and may contain complex temporal patterns that simpler prediction methods cannot fully capture.				
9		Find the article and full bibliographic details					
10	A Comparative Study of Statistical and Machine Learning Models on Carbon Dioxide Emissions Prediction of China (2023)	Research article indexed by PubMed (2023). Read the study record			Comparison of six models: Grey Model GM(1,1), ARIMA, SARIMAX, Artificial Neural Network, Random Forest, and LSTM; evaluation used MSE, RMSE, MAE, MAPE, and \ (R^2\)		[Future dEvaluate the models with newer data, examine performance during unusual emission changes, and test whether the best model generalizes to other regions. irection]

3. Methodology

3.1 Proposed System Flow Diagram

Insert the proposed-system flow diagram here as perf your project.

Recommended flow: Dataset → Data Cleaning → Missing-Value Treatment → Encoding → Feature Engineering/ Selection → Train/Test Split → Model Training → Hyperparameter Tuning → Prediction → Evaluation → Final Model.

1. Dataset Collection

Historical carbon emissions and relevant input features

2. Data Inspection and Cleaning

Check data types, duplicates, missing values, and inconsistencies

3. Exploratory Data Analysis

Study distributions, correlations, trends, and outliers

4. Feature Preparation

Select relevant features and encode categorical variables if needed

5. Train-Test Split

Separate training data from unseen testing data

6A. Linear Regression

Baseline model

6B. Gradient Boosting

Ensemble model

7. Model Evaluation and Comparison

MAE, MSE, RMSE, and R^2

8. Final Model and Prediction

Select the model based on test performance and generate emission estimates

3.2 Dataset Description

The project uses a carbon emission dataset containing historical emission values and relevant features that may influence emissions. Depending on the selected dataset, these features may include year, energy consumption, fuel usage, population, economic activity, or other environmental variables. The target variable represents the carbon emission value to be predicted. The dataset will be inspected to determine its source, number of records, number of features, data types, missing values, and measurement units. The exact dataset name, source URL, record count, and target-column name must be added after the dataset has been selected and examined.

3.3 Experimental Setup

[Hardware, OS, Python version, IDE/Jupyter/Kaggle environment, libraries, and versions.]

Operating system -macOS, Windows, or Linux

Programming language -Python

Development environment -Jupyter Notebook, Google Colab, or VS Code

Data processing -Pandas and NumPy

Visualization -Matplotlib and Seaborn

Machine learning -Scikit-learn

Models -Linear Regression and Gradient Boosting Regressor

Input data -Carbon emission dataset in CSV format

Evaluation -MAE, MSE, RMSE, and \(R^2\)

3.4 Data Preprocessing

Data preprocessing prepares the dataset for machine learning and reduces the risk of misleading results.

1. Data inspection: Examine column names, data types, dimensions, and summary statistics.
2. Duplicate removal: Identify and remove duplicate observations when they represent repeated records rather than legitimate measurements.
3. Missing-value treatment: Measure missingness and use an appropriate imputation strategy or remove unsuitable records when justified.
4. Outlier analysis: Identify unusually large or small observations and determine whether they are errors or genuine emission events.
5. Categorical encoding: Convert categorical input features into numerical form when required.
6. Numerical scaling: Apply scaling when useful for the selected algorithm. Standardization is not normally essential for ordinary Linear Regression or tree-based Gradient Boosting.
7. Target preparation: Verify that the emission target is numerical and expressed in consistent units.
8. Train-test splitting: Separate training and testing data before fitting the models. For time-dependent prediction, use a chronological split where appropriate.
9. Data leakage prevention: Fit imputers, encoders, feature selectors, and other learned preprocessing steps using training data only. Do not use future emission values to predict past values.

3.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the characteristics of the dataset before model training. Summary statistics help identify the average, spread, minimum, and maximum values of emission measurements. Histograms can show the distribution of carbon emissions, while scatter plots can reveal relationships between emissions and individual input features. Correlation analysis can identify linear associations between numerical variables. Box plots can help examine extreme

observations, and line plots can reveal changes over time when the dataset contains a time feature. These visualizations help guide preprocessing and feature selection without assuming that correlation proves causation.

3.6 Feature Engineering and Feature Selection (if applicable)

Feature engineering involves preparing useful input variables for the prediction models. Depending on the available data, numerical variables may be transformed, relevant categorical variables may be encoded, and justified features such as energy consumption per capita may be derived from existing columns. Feature selection should be guided by domain relevance, exploratory analysis, and validation performance. Features that directly reveal the target value or contain information unavailable at prediction time should be excluded. The final feature list and the reasons for selecting or removing individual variables should be documented after examining the dataset.

3.7 Model Building and Training

Two regression algorithms are implemented in this project.

Linear Regression: Linear Regression estimates the relationship between input variables and a numerical target. It learns coefficients that minimize the sum of squared prediction errors on the training data. It provides a relatively simple baseline and can help explain the direction and magnitude of linear relationships, subject to the model's assumptions.

Gradient Boosting Regression: Gradient Boosting builds an ensemble of decision trees sequentially. Each additional tree is trained to improve the predictions of the existing ensemble by reducing the selected loss function. This allows the model to learn nonlinear relationships and interactions among features. However, its complexity must be controlled to reduce overfitting.

Both models should be trained using the same training data and evaluated on the same test set. A validation procedure or cross-validation can be used to select parameters without repeatedly optimizing against the test set.

3.8 Hyperparameter Tuning(if applicable)

Hyperparameter tuning can improve the performance of Gradient Boosting Regression. Parameters worth investigating include the number of boosting stages (`n_estimators`), learning rate (`learning_rate`), maximum tree depth (`max_depth`), and the minimum number of samples required for a split or leaf. Grid Search or Randomized Search with cross-validation can be used to select suitable values. The search should use only the training data, and the final model should be evaluated on the held-out test set. Linear Regression can be retained as the baseline; additional regularization can be considered if justified

3.9 Model Evaluation

The project uses four regression evaluation metrics.

10. Mean Absolute Error (MAE): Measures the average absolute difference between actual and predicted emissions. Lower values indicate smaller average errors.

11. Mean Squared Error (MSE): Calculates the average squared prediction error. It penalizes larger errors more strongly.

12. Root Mean Squared Error (RMSE): The square root of MSE, expressed in the same units as the target variable.

13. Coefficient of Determination (\(R^2\)): Measures how well the model explains variation in the target relative to a mean-based baseline. Higher values generally indicate better fit, but \(R^2\) can be negative on test data.

The preferred model should generally have lower MAE and RMSE and a higher \(R^2\), considered together. Comparisons must use the same test set and target units.

4. Results and Discussion

4.1 Preprocessing Results

The dataset was inspected to determine its initial dimensions, feature types, missing values, and duplicate records. Missing values were treated using the selected preprocessing strategy, and duplicate records were examined before removal. Relevant features were prepared for regression analysis, and the dataset was divided into training and testing subsets. The final number of observations and features was recorded after preprocessing.4.2 Model Performance.

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**ROC AUC**
Existing study	**0.73**	**0.76**	**0.73**	**0.68**	**0.73**
Decision Tree	0.93	0.96	0.96	0.96	0.96
Random Forest	0.86	0.91	0.89	0.89	0.89
XGBoost	0.91	0.89	0.85	0.84	0.85
Gradient Boosting	0.85	0.89	0.86	0.85	0.86

4.3 Visual Results and discussion

The following visualizations are recommended for the final report:

1. Emission distribution: A histogram showing the distribution of target values.

2. Correlation heatmap: A visualization of relationships among numerical variables.

3. Actual versus predicted plot: A scatter plot comparing actual test values with model predictions. Predictions close to the diagonal indicate better agreement.

4. Residual plot: A plot of prediction errors that can reveal systematic errors or changing error variance.

5. Model comparison chart: A bar chart comparing MAE or RMSE for Linear Regression and Gradient Boosting.

6. Feature importance plot: A feature-importance chart for Gradient Boosting, if appropriate. Feature importance indicates predictive contribution, not necessarily causation.

The discussion should explain which model performed better on the test data, how large its prediction errors were, whether residuals show systematic patterns, and whether the results appear consistent with the intended application. Any claims about improved accuracy or better generalization must be supported by the measured results.

5. Conclusion

This project focuses on developing a machine learning-based system for predicting carbon emission levels using Linear Regression and Gradient Boosting Regression. The proposed methodology includes dataset preparation, data preprocessing, exploratory data analysis, feature selection, model training, and evaluation. The two models are compared using MAE, MSE, RMSE, and \(R^2\) to assess their predictive performance. Linear Regression provides a baseline for understanding relationships between input features and emissions, while Gradient Boosting can capture more complex patterns in the data. The final conclusion regarding the better-performing model will be based on the experimental results obtained from the selected dataset. The project demonstrates how machine learning can contribute to carbon emission analysis and support data-informed environmental planning.

6. Future Work

The project can be extended in the following ways:

1. Larger datasets: Include more historical observations and data from different regions and industries.

2. Additional algorithms: Compare Gradient Boosting and Linear Regression with Random Forest, XGBoost, and other suitable regression methods.

3. Advanced optimization: Apply systematic hyperparameter tuning and cross-validation to improve model selection.

4. Explainable AI: Use appropriate feature-importance methods or SHAP to better understand the factors influencing predictions.

5. Time-series forecasting: Incorporate temporal models when the goal is to predict future emissions over successive periods.

6. Real-time data integration: Integrate energy consumption or industrial monitoring data where reliable data sources are available.

7. Web-based application: Develop a user interface that accepts relevant input values and displays predicted emissions alongside explanatory visualizations.

8. External validation: Evaluate the selected model on independent datasets to investigate how well it generalizes.

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